Andrew Hinh

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Education

University of California, Los Angeles Expected 06/2028
Bachelor of Science in Electrical Engineering

Work Experience

Embedded Systems Intern, Flight Interactive System Technologies August 2026 – Present
NASA Armstrong Flight Research Center (AFRC) *Edwards AFB, CA*

- Troubleshot and resolved a magnet separation-distance failure mode during twist-to-release testing through root-cause analysis of test data, increasing holding force by up to 200 lbs and improving product reliability for larger UAV reconnection testing
- Developing long-range RF mesh radio test architecture linking multiple UAVs and a ground station over 4 km, validating relative navigation data exchange across varying antenna gain and bandwidth conditions
- Performed statistical analysis of LiDAR and computer vision test data across 15 flight tests on a Raspberry Pi 5, supporting data-driven troubleshooting for precise UAV alignment
- Designed power distribution hardware using 12V-to-5V DC-DC buck converters, ensuring regulated power delivery to sensor systems

Undergraduate Research Assistant April 2026 – Present
Power Electronics and Energy Control Laboratory at UCLA *Los Angeles, CA*

- Built a Python/PyVISA-based test automation GUI integrating six lab instruments (oscilloscope, DMM, e-load, PSU, PWM driver, thermal cameras), enabling consistent statistical data and achieving 85% efficiency sweep for step down synchronous buck converters
- Documented and standardized remote test procedures into a simple GUI, supporting repeatable remote testing for 30+ students

Electronics/Avionics Team Lead Sep. 2025 – Present
UAS (Uncrewed Aerial Systems at UCLA) *Los Angeles, CA*

- Led PCB development for a sub team of 4 members for an eVTOL power distribution board from component selection and schematic capture through PCB layout, architecting passthrough and step down voltage delivery from batteries to ESCs and motors
- Developed test procedures and acceptance criteria for a custom IMU, performing board troubleshooting and validation testing on a schematic integrating an accelerometer, magnetometer, gyroscope, and microcontroller in KiCad to ensure product integrity before flight deployment
- Built electronic test benches and fixtures, performing crimping and soldering of electrical components to produce flight-ready wiring harnesses and maintained build documentation to support repeatable assembly across the test fleet
- Supported mission software planning and configured live telemetry radio links to relay MAVLink messages between the ground station and flight controller

Lead Mechanical Design and Outreach team Sep. 2021 – Mar. 2025
First Tech Challenge Robotics, 21980 Bread Pandas *Fountain Valley, CA*

- Led systems integration and task delegation for two subteams, helping raise $10,000 in sponsorships and achieve a top 2% regional rank
- Designed, prototyped, and optimized mechanical components using engineering principles like constant spring force and servo torque estimations

Projects

IMU NeckBrace PCB Design | *MPU 6050, JST Connection, I2C, ESP32, Altium, PCB Layout, Schematic Capture, C++*

- Designed a schematic for an MPU-6050-based IMU breakout board supporting real-time neck position sensing that corrects cervical misalignment via 12 independently actuated pneumatic solenoid valves
- Specified I2C bus circuitry (4.7kΩ pull-up resistors), power supply decoupling, and JST cable interface to main control board to allow for easy mechanical design along with custom Jansen linkage mounting position

Vertical Take-Off Landing (VTOL) Drone Design | *Fusion360, ArduPilot, Pixhawk, ESCs, Battery Selection, RC Systems*

- Designed a vertical take-off and landing (VTOL) drone, overseeing CAD modeling in Fusion360 and full electrical system integration. Utilizing 0.8mm 400mm carbon fiber rods for the main drone frame, along with 3D printed drone architecture for electronic housing/placement
- Configured and tuned a Pixhawk flight controller running ArduPilot firmware, integrating ESCs, motors, and sensor feedback for stable transition between vertical and forward flight while ensuring live transmitting of video/feed through telemetry radio

Technical Skills

Languages: C++, Java, Python, MATLAB, McCreator, Roblox Studio
Hardware: Oscilloscope, Digital Multimeter, Electronic Load, DC Power Supply, Waveform Generator, AD2
Design & Tools: Circuit Design, PCB/PCBA Layout & Design, Schematic Capture, Test Procedure Development, Root-Cause Troubleshooting, Technical Documentation, Control Systems, LTspice, KiCad, Altium, Soldering, STM32, Arduino
Mechanical Engineering: Creo, Fusion360, 3D-printing, CNC Manufacturing, Rapid Prototyping, CAM, Drill Press